

Build an AI Agent with SPARK

Turn your flows into tools, wrap them in a chat agent, and let it decide which tool to call from a plain-English request.

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

Open SPARK and set an API key in Settings (gear, top-right). The activity slide shows the agent on gemini-2.5-flash but you can use a GPT model (Azure OpenAI / gpt-5.4-nano) - if Gemini is busy on your end (check the announcement for the endpoint and API key to configure the AI provider). The files used are automated-formulation.pdf and arsenal-football.jpg.

Tool	SPARK - Studio for Prototyping Agent Reasoning & Knowledge
Example	Conversational Agent Demo
Tools (flows)	PDF Translator (PDF -> Vietnamese) and Object Detector (find + box an object)
What you build	A chat agent that picks the right flow to answer each request
Key idea	A flow on the canvas becomes a tool the agent can call - just like ChatGPT plugins

Step 1 - Open the Conversational Agent Demo

An AI agent is an LLM that can decide to call tools. In SPARK, each flow you build can become a tool. This example ships with two flows on one canvas, ready to be turned into tools.

1. On the Gallery, open the Conversational Agent Demo (Confirm to clear the canvas).
2. Flow 1 (top): Upload Document -> Translate to Vietnamese -> Show Translation.
3. Flow 2 (bottom): Upload Image + Object to Find -> Find Object AI -> Combine for Viewer -> Display Result.
4. Open the Flow Control Panel to see each flow's name and description - this text is what tells the agent what each tool does.

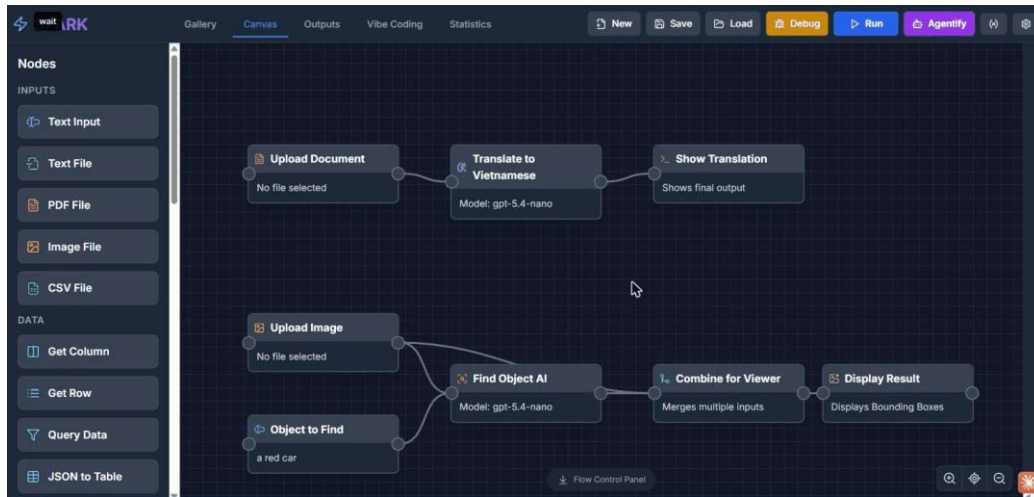


Figure 1 - Two flows on one canvas: a PDF translator (top) and an object detector (bottom). Each will become a tool the agent can call.

Step 2 - Agentify: wrap the flows in an agent

Click Agentify (top-right) to open the Agent Configuration dialog, then set up a new agent:

5. Agent Name: My Agent. Agent System Prompt: leave the default "You are a helpful AI assistant."
6. AI Provider / Model: here we use Azure OpenAI / gpt-5.4-nano.
7. Tick Display full tool output in chat (so tool results show right in the chat), Enable Planning Mode, and Enable Action Mode (so the agent can use flows as tools).
8. Under Available Flows (Tools), tick both PDF Translator and Object Detector.
9. Click Create Agent.

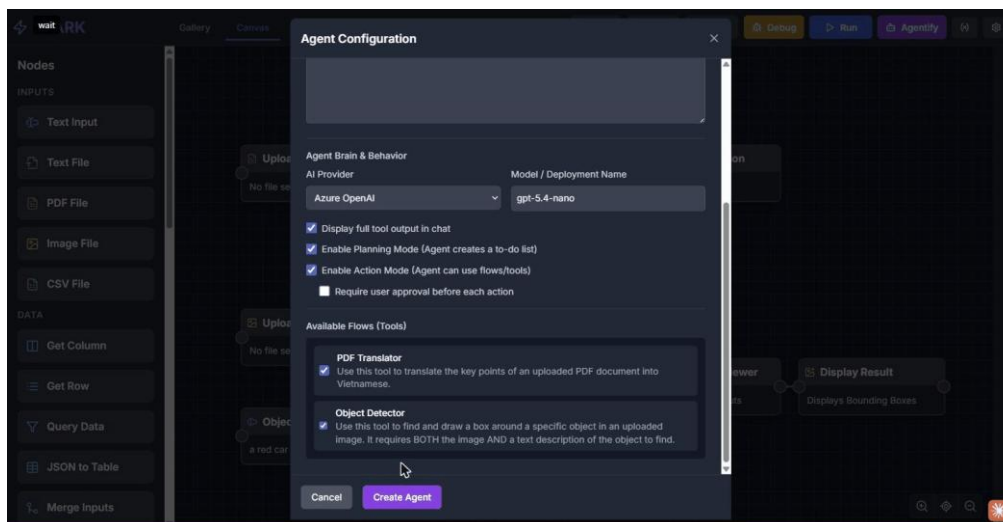


Figure 2 - The Agent Configuration: Planning + Action modes on, full tool output on, and both flows ticked as tools.

Tip: "Require user approval before each action" is left unchecked here so the agent runs tools automatically. Tick it if you want to approve each tool call first.

Step 3 - Test the translation tool

Once created, the agent chat opens (the purple chat icon, top-right). It works like ChatGPT or Val - but it can call your flows.

10. Click the paperclip in the chat and attach automated-formulation.pdf.
11. Type: "Help me translate the key points from this attached paper" and send.
12. The agent recognises this needs the PDF Translator tool, runs that flow (you'll see it light up on the canvas), and - because Display full tool output is on - prints the full Vietnamese translation right in the chat.

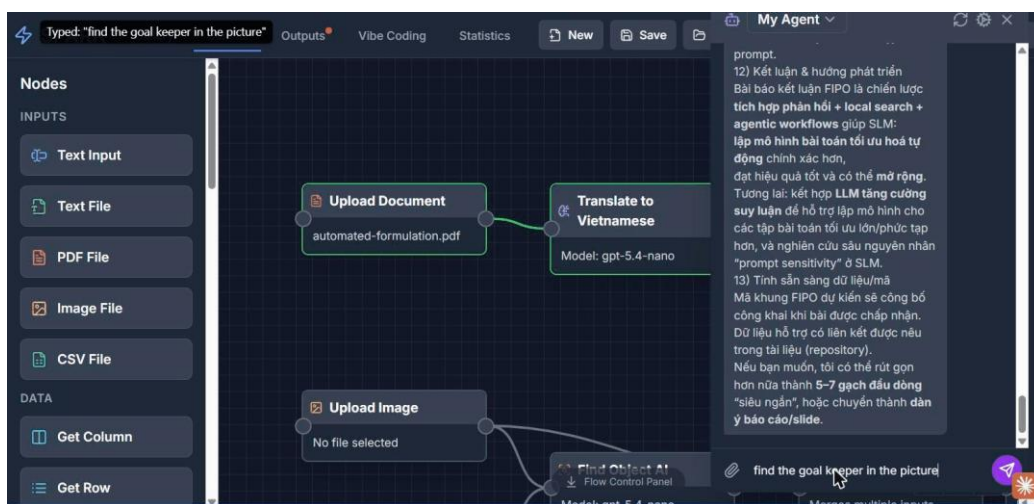


Figure 3 - The agent chose the PDF Translator tool and returned the document's key points in Vietnamese, shown in the chat.

Step 4 - Test the object-detection tool

The same agent can switch tools based on what you ask. Attach an image and ask it to find something.

13. Attach arsenal-football.jpg via the paperclip.
14. Type: "find the goal keeper in the picture" and send.
15. The agent now calls the Object Detector tool instead, runs that flow, and tells you to click the Display Result node. Click it to open the image with a labelled box around the goalkeeper.

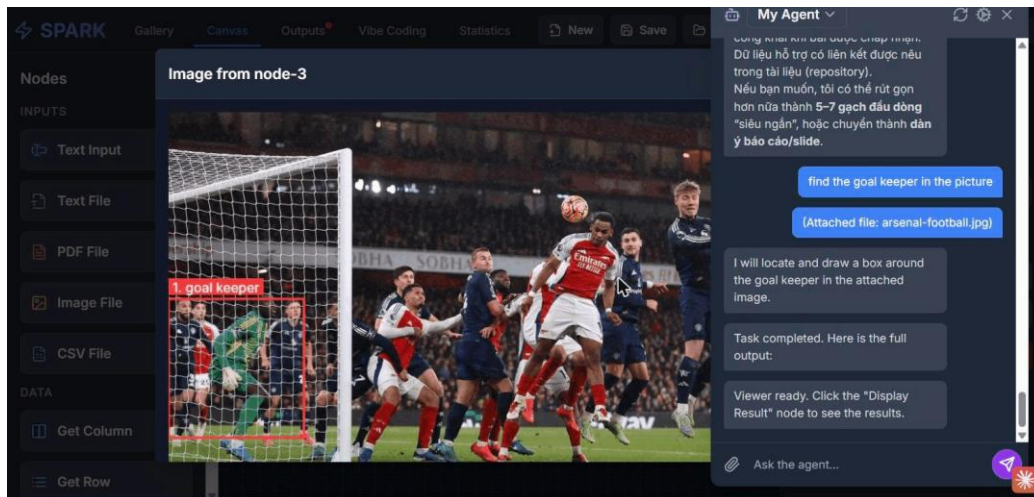


Figure 4 - Same agent, different request: it called the Object Detector tool and drew a box around the goalkeeper.

This is the essence of an AI agent: you describe what you want in plain language, and the agent decides which tool (flow) to use, runs it, and reports back - no need to open the flow yourself.

Good to know

- A flow becomes a tool: the agent reads each flow's name and description to decide when to use it, so write those clearly.
- The agent picks the tool from your wording - "translate this PDF" vs "find the goalkeeper" route to different flows automatically.
- Object detection can still be flaky; if a tool call errors, ask again or re-run - the agent will retry the flow.